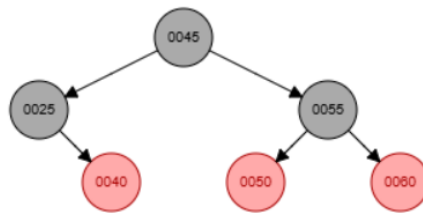


Red/Black Tree

 Insert Delete Find Print☐ Show Null Leaves

After inserting 45, 35, 55, 25, 40, 50 and 60, the Red-Black Tree is balanced using recolouring when a Red-Red violation occurs. When 25 is inserted, both its parent 35 and uncle 55 are RED, so they are recoloured BLACK. After inserting the remaining nodes, the tree satisfies all Red-Black properties. When node 35 is deleted, its successor 40 is used to maintain the BST structure. The required recolouring maintains the black-height property. No rotation is required in this particular deletion. The final Red-Black Tree has 45 as the BLACK root, 25 and 55 as BLACK children, and 40, 50 and 60 as RED nodes.